



High-temperature wettability and film-forming behavior of borosilicate glass lubricant on Ti6Al4V substrate



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Abstract

Ti6Al4V alloy is prone to oxidation and surface damage during high-temperature die forging, thus effective thermal protection is critically required. Owing to their excellent high-temperature stability, favorable lubricity, environmental friendliness, and low cost, glass lubricants have become an ideal protective material for Ti6Al4V alloy in high-temperature die forging processes. This study focuses on a borosilicate glass lubricant, employing high-temperature dynamic observation and multi-scale characterization to systematically investigate the evolution of its structure, morphology, interfacial behavior, and wetting characteristics within the temperature range of 550–950 °C. The results indicate that the glass lubricant undergoes stage-wise transformations with increasing temperature: it begins to melt and forms a porous structure at 650 °C; a continuous film is achieved by 750 °C; at 950 °C, the melt exhibits significantly improved wettability on the Ti6Al4V alloy surface. Based on these

dynamic evolution behaviors, the wetting mechanism and film formation process are further elucidated. The findings regarding the high-temperature behavior of the borosilicate glass lubricant provide important theoretical support for improving the high-temperature protective performance of Ti6Al4V alloy.



Keywords

Borosilicate glass lubricant; High-temperature wettability; Film formation; Ti6Al4V alloy

1. Introduction

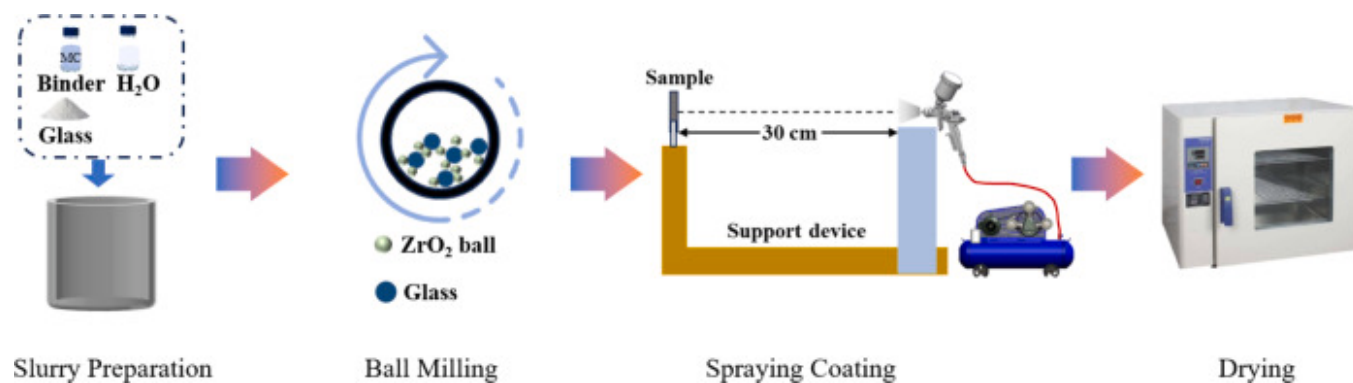
Ti6Al4V(TC4) alloy has been widely employed in aerospace, automotive, medical, and chemical industries due to its advantageous combination of lightweight characteristics, high strength, exceptional corrosion resistance, and favorable tensile and toughness properties [1], [2], [3], [4], [5]. Prior to service implementation in aviation applications, TC4 alloy components need to undergo hot processing techniques such as high-temperature die forging and hot extrusion to achieve refined microstructures, specific geometries, and enhanced mechanical performance. During the hot processing of TC4 alloy, lubricants are typically applied to provide high-temperature surface protection [6], [7]. Glass-based coatings have emerged as an ideal lubrication material for TC4 alloy die forging processes owing to their superior wear resistance, non-polluting nature, process simplicity, and cost-effectiveness [8], [9], [10]. Furthermore, the glass lubricant has a dense structure and high-temperature stability, which can effectively prevent the surface oxidation of TC4 alloy at high temperatures. Therefore, it has attracted extensive attention from researchers at home and abroad [11], [12], [13].

During high-temperature die forging of TC4 alloy, before the TC4 alloy undergoes severe oxidation, the glass lubricant undergoes sequential softening and gradual melting [8], [14]. This molten phase subsequently spreads across the substrate surface, forming a dense vitreous film that establishes robust adhesion with the substrate. The resulting continuous barrier effectively impedes oxygen diffusion into the TC4 alloy matrix, thereby substantially enhancing its high-temperature oxidation resistance [15], [16],

[17]. Simultaneously, within the operational temperature range of 800–950 °C, the glass lubricant develops a fluid-sealed interfacial layer with optimal flow characteristics. This layer facilitates efficient fluid lubrication, significantly reducing interfacial friction between the forging component and die surfaces while achieving synergistic protective and lubricative effects [18], [19], [20], [21]. The exceptional lubrication performance and protective efficacy of the glass lubricant fundamentally derive from its superior wettability and film-forming capacity on TC4 alloy substrates [22], [23], [24]. Consequently, systematic investigation of the wetting behavior and spreading mechanisms governing glass lubricants on TC4 alloy surfaces holds critical significance for advancing thermal protection technologies in high-temperature forming processes.

Glass lubricants are of substantial engineering importance in hot metal forming because they can deliver both surface protection and interfacial friction reduction at elevated temperatures, thereby helping maintain dimensional accuracy, decrease energy consumption, and extend die life [25], [26], [27]. Since the late 1950s, extensive efforts have been devoted to glass and silicate based protective lubricating coatings, which have been progressively implemented in extrusion, die forging, and rolling, providing effective oxidation resistance, decarburization mitigation, and lubrication [28], [29], [30], [31]. A broad consensus has emerged that the efficacy of glass lubricants is strongly dependent on the processing temperature window. Only when the softening point and high temperature rheological response are matched to the processing temperature range can the glass form a continuous and stable molten viscous film during thermal forming, enabling the concurrent functions of an oxidation barrier and liquid phase lubrication [32], [33], [34], [35]. In recent years, systematic studies have advanced both application oriented validation and mechanistic understanding. Matsumoto et al. [25] showed that tribological effectiveness exhibits pronounced temperature dependence. Fan et al. [36] reported a borosilicate glass lubricant that forms a viscoelastic film at 800 °C during hot extrusion of industrially pure titanium, markedly reducing frictional losses, oxide layer thickness, and production costs. Tang et al. [37] systematically evaluated oxidation and hot corrosion resistance, highlighting that improved wettability and interfacial bonding are essential for suppressing crack propagation and film delamination. Ji et al. [38] used high temperature contact angle measurements to demonstrate that pre oxidation parameters modify the surface topography and chemistry of TC4 alloy, thereby significantly altering the wetting behavior of the glass. Overall, current research on glass lubricants for high temperature heat treatment of titanium alloys has primarily focused on enhancing oxidation resistance and optimizing lubrication performance. However, systematic investigations into glass and titanium interfacial wettability remain limited, particularly regarding the temperature gradient driven film forming diffusion processes of glass lubricants across titanium surfaces, which warrants further elucidation.

This study focuses on borosilicate glass lubricants conventionally utilized in TC4 alloy high-temperature die forging. We dynamically captured the wetting and spreading processes of glass lubricants on TC4 alloy surfaces. A systematic analysis was conducted to investigate the structural evolution and microstructural transformations of glass lubricants during progressive thermal elevation. Moreover, this study further explored the wetting behavior of borosilicate glass on the TC4 alloy surface, the diffusion behavior of the glass melt on the TC4 alloy surface, and the film-forming mechanism of the glass lubricant on the TC4 alloy surface. The research findings provide theoretical guidance for regulating the high-temperature protective performance of the glass through the adjustment of surface microstructure and surface wettability.



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Fig. 1. Process flow diagram for the preparation of glass lubricant.

2. Method

2.1. TC4 alloy

The present investigation employed TC4 alloy as the metallic substrate. Specimens with dimensions of 20 mm × 20 mm × 5 mm were prepared through a wire-cutting machine. To eliminate surface oxide layers and contaminants while enhancing surface roughness for improved glass lubricant adhesion between the glass lubricant and substrate, all samples were mechanically ground using 400-grit SiC abrasive paper. Subsequently, the processed TC4 alloy specimens were immersed in a mixed solution of anhydrous ethanol and acetone (1:1 v/v) and subjected to ultrasonic cleaning for 30 min to achieve thorough degreasing and

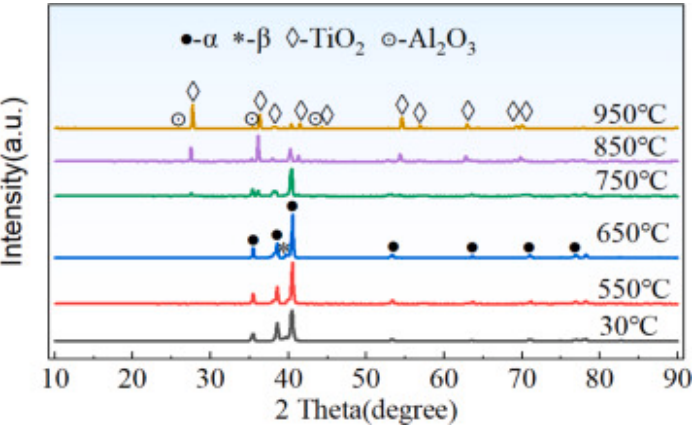
purification. Finally, the degreased specimens were dried in a constant-temperature drying oven at 80 °C for 2 h to ensure complete surface dehydration, thereby preparing them for subsequent slurry of glass powder.

2.2. Glass powder preparation

The glass frit was prepared via a conventional water-quenching method. Following the glass formulation specified in Table 1, analytical-grade reagents (Sinopharm Chemical Reagent Co., Ltd.) were precisely weighed and transferred to a ball milling jar. Anhydrous ethanol was introduced as a grinding aid (liquid-to-powder ratio 2:1 v/w), followed by mechanical homogenization at 300 rpm for 4 h to achieve optimal raw material uniformity. The mixture was subsequently dried and loaded into an alumina crucible, which was then heated in an electric resistance furnace at 1500 °C for 2 h to ensure complete vitrification and homogeneous melt formation. The crucible was rapidly removed and quenched in deionized water at ambient temperature to obtain the glass frit. The resultant frit was subjected to secondary wet ball milling with anhydrous ethanol at 400 rpm for 8 h to achieve particle refinement, followed by final drying in an oven for subsequent glass lubricant preparation.

Table 1. The composition of glass powder(wt%).

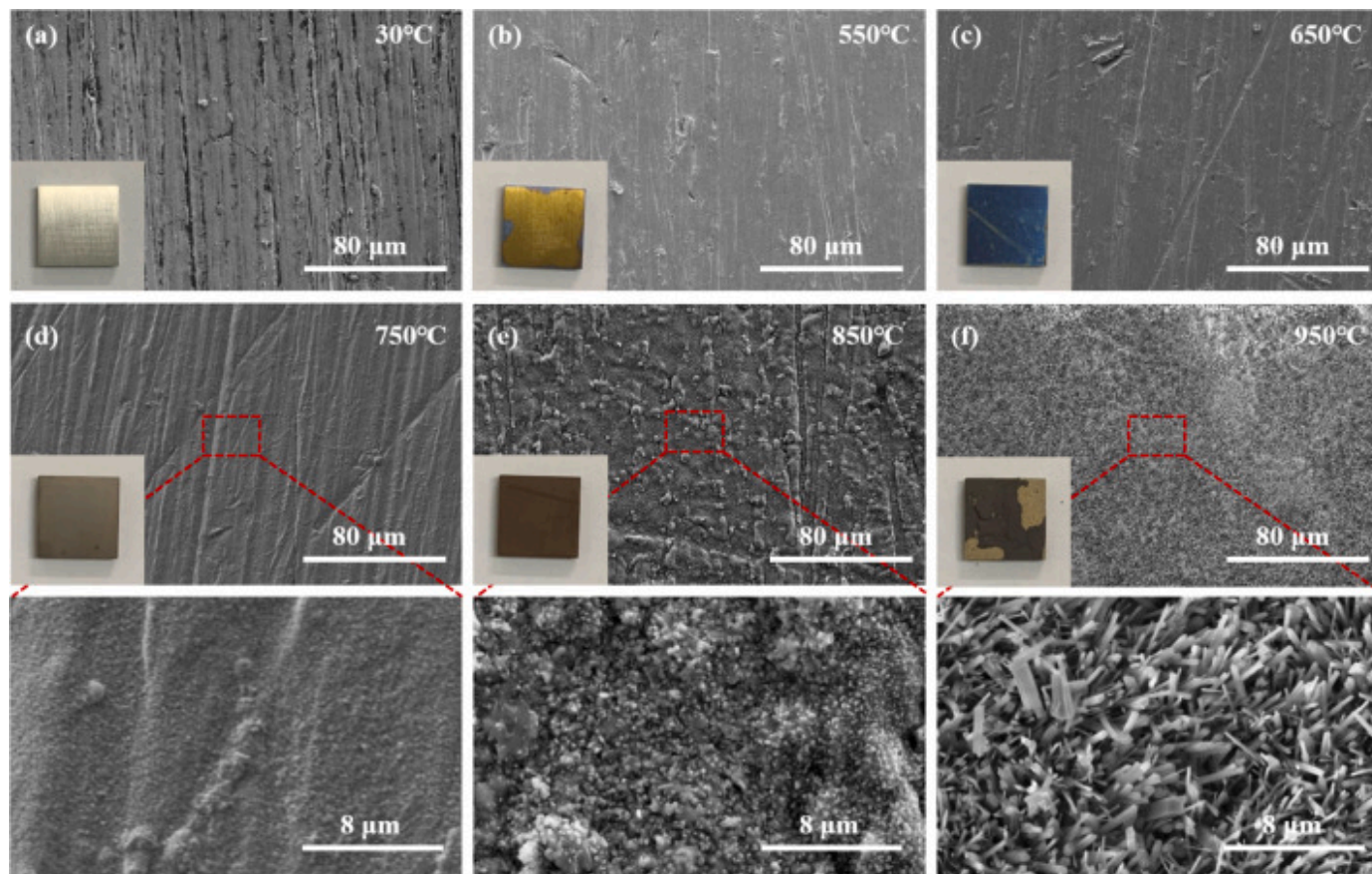
SiO ₂	Al ₂ O ₃	B ₂ O ₃	CaO	Na ₂ O	Li ₂ O	MgO	BaO	ZnO
50	7	20	6	7	2	2	4	2



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Fig. 2. XRD patterns of TC4 alloy substrate with increasing temperature.



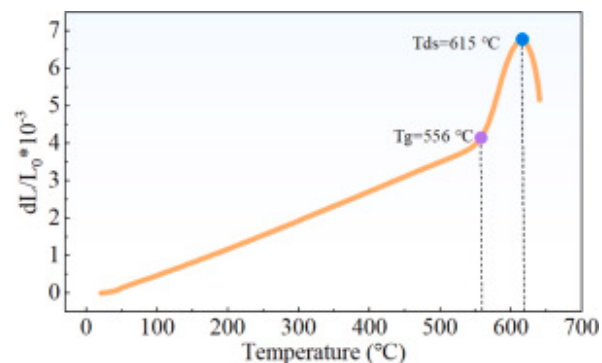
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Fig. 3. Macroscopic and microscopic morphological evolution of TC4 alloy substrate surface with increasing temperature: (a) 30 °C, (b) 550 °C, (c) 650 °C, (d) 750 °C, (e) 850 °C, (f) 950 °C. (The illustration is the corresponding optical photograph.)

2.3. Glass lubricant preparation

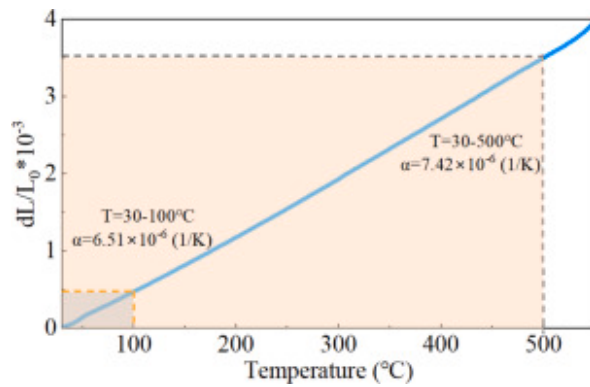
The glass frits, carboxymethyl cellulose (CMC), and deionized water were homogenized in a 4:2:4 mass ratio using a planetary ball mill operated at 400 rpm for 4 h to prepare glass slurry. The spray gun was then connected to an air compressor calibrated to deliver a consistent spray pressure of 0.2 MPa to achieve homogeneous atomization. Pre-treated TC4 alloy specimens were positioned 30 cm from the spray nozzle, and all six surfaces of the pre-dried substrates were manually coated through sequential spraying passes to achieve full glass layer deposition. Post-spraying specimens were transferred to a drying oven for 12 h at 80 °C, followed by heat treatment in a muffle furnace under atmospheric conditions according to preprogrammed temperature–time profiles. Fig. 1 is a flow chart for the preparation of glass lubricants.



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(a).

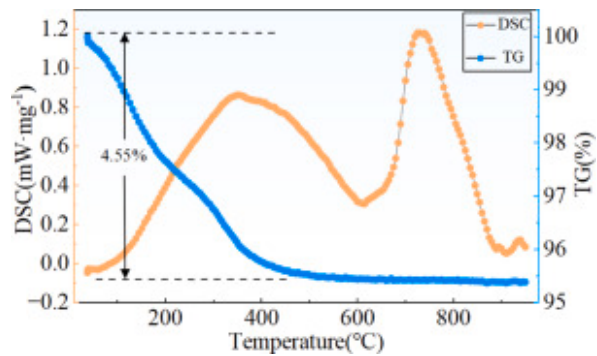


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(b)..

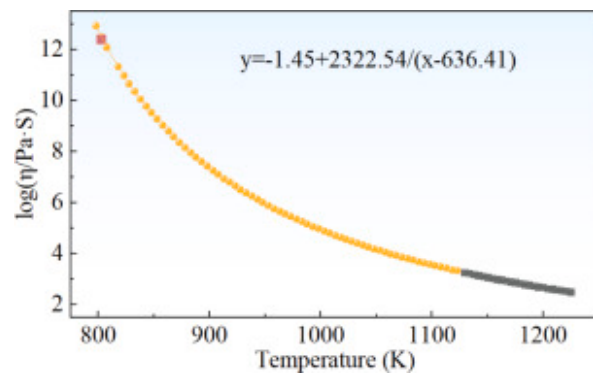
Fig. 4. (a) Thermal expansion softening temperature of the glass lubricant. (b) Thermal expansion coefficient of glass lubricant.



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(a)..

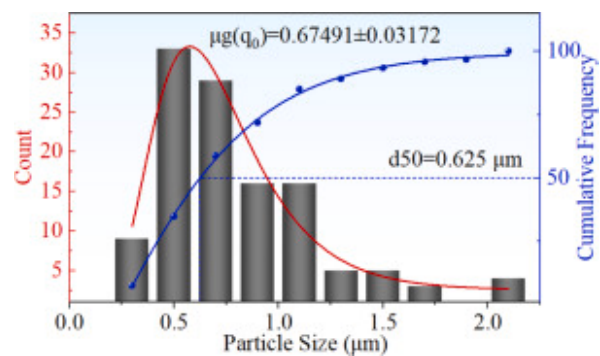


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(b)..

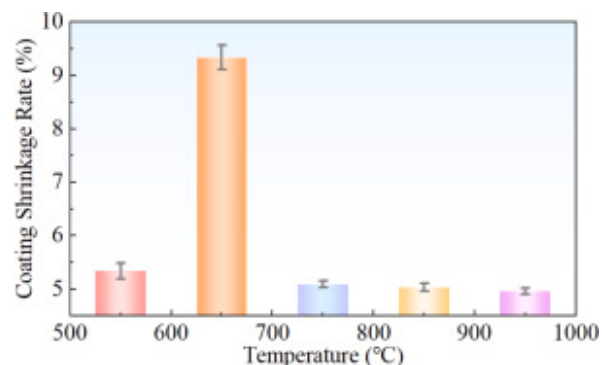
Fig. 5. (a) TG-DSC curve of the glass lubricant. (b) The curve of the viscosity of glass lubricant varying with temperature.



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(a)..



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(b)..

Fig. 6. (a) Particle size analysis of glass frit. (b) shrinkage rate of glass lubricants after heat treatment at different temperatures.

2.4. Characterization methods

Microstructural characterization was performed using a scanning electron microscope (SEM, Hitachi TM4000Plus, Japan), with compositional distribution determined by energy-dispersive X-ray spectroscopy (EDS, Bruker QUANTAX, Germany). Phase composition analysis of the original samples, including thermal oxidation-induced phase transitions in TC4 alloy and glass lubricants, was conducted via X-ray diffraction (XRD) (Rigaku D/max 2500 PC, Japan, Cu K α , λ = 0.1548 nm) at a scanning rate of 0.05°/step and an operating voltage of 40 kV. Dynamic Wetting angles on thermally oxidized metal surfaces were quantified using a contact angle analyzer (OCA25-HTV1800, DataPhysics Instruments, Germany) under controlled oxidation conditions. The high-temperature viscosity test was conducted using a high-temperature rotational viscometer (RSV-1600, Orton, USA). Thermal analysis was performed using a NETZSCH STA449 differential scanning calorimeter under an air atmosphere, with a heating rate of 10 K/min over the temperature range from room temperature to 1000 °C. The coefficient of thermal expansion was measured via a NETZSCH DIL 402 PC dilatometer, with a heating rate of 5 K/min over the temperature range from room temperature to thermal expansion softening point. The frictional behaviors of nickel-based superalloy 4169 and TC4 alloy lubricated with glass lubricants were investigated using a pin-on-disk tribometer (Model: GF-1) under a high-temperature environment of 900 °C in air atmosphere. The diameter of the pin was 6 mm. The tests were conducted under the conditions of a motor speed of 5 rpm, an

applied load of 15 N, and a reciprocating sliding distance of 5 mm. Each test was repeated at least three times to verify the reliability of the results.

3. Result

3.1. The effect of temperature on the phase and surface morphology of TC4 alloy

The TC4 alloy exhibits extremely high chemical reactivity and is prone to oxidation by oxygen and water vapor at high temperatures. Fig. 2 presents the XRD patterns of TC4 alloy oxidized at different temperatures. At 550 °C and 650 °C with short oxidation durations, the limited oxide formation resulted in undetectable diffraction peaks of oxidation products. Upon reaching 750 °C, distinct diffraction peaks corresponding to TiO_2 and Al_2O_3 emerged, indicating substantial oxidation consistent with previous reports. Progressive intensification of oxidation occurred as temperatures increased to 850 °C and 900 °C, manifested by: (1) increased quantity and intensity of TiO_2 ($2\theta = 27.4^\circ, 36.1^\circ$) and Al_2O_3 ($2\theta = 35.1^\circ, 43.4^\circ$) diffraction peaks, (2) corresponding attenuation of TC4 alloy peaks ($2\theta = 38.4^\circ, 40.2^\circ$).

The surface properties of the TC4 alloy have a significant impact on the wettability and film-forming ability of borosilicate. Therefore, we collected optical photographs of the TC4 alloy as the oxidation temperature changed in the air, with the oxidation time of each group being 30 min, as shown in Fig. 3. The unoxidized TC4 alloy appears silver-gray. The surface of TC4 alloy after thermal oxidation exhibits different colors, and the differences in surface color may be related to the formation, distribution, and content of oxidation products. This chromatic evolution enables preliminary assessment of surface oxide composition and film thickness in TC4 alloy through visual inspection. When the TC4 alloy is heated to 550 °C, the surface of the substrate turns golden-yellow with localized blue edges, consistent with Zhang et al.'s identification of golden-yellow TiO and blue coloration corresponding to thin TiO_2 layers (50–100 nm thickness) [38], [39]. Upon reaching 650 °C, complete surface conversion to uniform blue coloration occurred, indicative of coherent TiO_2 layer development. Progressive temperature escalation induced sequential color transitions: dark gray (750 °C), reddish-brown (850 °C), and brown (900 °C), corresponding to oxide layer thickening and structural densification. Notably, partial spallation of the oxide scale was observed post-oxidation at 950 °C, suggesting thermal stress-induced delamination beyond critical oxide thickness.

A further analysis was conducted on the changes in the surface micro-morphology of the samples as the temperature increased. Fig. 3 shows the SEM images of the TC4 alloy at different temperatures. In Fig. 3(a), the scratches on the surface of the unoxidized

TC4 alloy were mainly caused by grinding with SiC sandpaper. With progressive thermal elevation, gradual oxidation obscures these surface striations (Fig. 3(b)–(f)). At 750 °C, small oxide particles (50–150 nm diameter) were clearly visible on the surface of TC4 alloy (Fig. 3(d)). Further temperature escalation to 850 °C induces substantial oxide proliferation, characterized by coarsened dimensions (300–500 nm) (Fig. 3(e)). Upon reaching 950 °C, complete surface coverage by columnar oxide grains (1.2–3.8 μm length, 0.3–0.7 μm width) is observed. This progression confirms intensified oxidation kinetics beyond 750 °C, where accelerated nucleation and growth of TiO_2 and Al_2O_3 particles collectively enhance surface roughness through cumulative oxide deposition.

3.2. Properties of glass lubricant

Fig. 4 shows the thermal expansion curve of the borosilicate glass lubricant. As shown in Fig. 4(a), when the temperature rises to 556 °C, the slope of the curve changes significantly with a noticeable inflection, indicating that the material begins to transition from a rigid, brittle glassy state to a viscoelastic rubbery state [40]. This temperature is referred to as the glass transition temperature. At 614.9 °C, the curve reaches its maximum value, which corresponds to the dilatometric softening point, marking the critical transition of the glass from a viscoelastic solid to a viscous fluid. At this point, the glass structure can no longer resist external forces and begins to undergo viscous flow under the slight pressure of the push rod and its own weight. Macroscopically, this is manifested as the expansion of the sample reaching its limit, followed by a sharp contraction. Beyond 614.9 °C, the curve declines rapidly, indicating accelerated contraction of the sample. At this stage, the glass has fully entered a softened state, and its deformation mechanism shifts from being dominated by thermal expansion to being governed by viscous flow. The dilatometric softening point of this glass lubricant (614.9 °C) is significantly lower than the onset temperature of severe oxidation for TC4 alloy substrate (approximately 750 °C). This implies that during the heating of the TC4 alloy billet to the forging temperature, the glass lubricant softens and melts in advance, forming a protective molten film before the die-forging operation begins. This film effectively isolates the TC4 alloy substrate surface from oxygen, thereby preventing oxidation and providing essential lubrication for the subsequent forging process.

As a temporary high-temperature protective coating for TC4 alloy, the high-temperature adhesion and low-temperature spallation characteristics of the glass lubricant are critical indicators for evaluating its suitability. To achieve effective thermal protection and facilitate easy removal after cooling, the lubricant is required to remain stably adhered at high temperatures while being capable of automatic spallation at low temperatures. According to the theory proposed by Harris, a thermal expansion mismatch exceeding 25% between the coating and the substrate enables automatic or light-tapping-induced spallation of the

coating [41]. As shown in Fig. 4(b), this study calculates the average coefficients of thermal expansion of the glass lubricant over the temperature ranges of 30–100 °C and 30–500 °C to be $6.51 \times 10^{-6} \text{ K}^{-1}$ and $7.42 \times 10^{-6} \text{ K}^{-1}$, respectively. Compared with the CTE of TC4 alloy, which ranges from $8.5\text{--}9.5 \times 10^{-6} \text{ K}^{-1}$ [42], [43], the thermal expansion mismatch between the glass lubricant and TC4 alloy is approximately 35% in the low-temperature range (30–100 °C), significantly exceeding the 25% debonding threshold. This indicates that the coating meets the conditions for automatic spallation upon cooling to low temperatures. In contrast, the thermal expansion mismatch in the high-temperature range (30–500 °C) is 18%, which is below the aforementioned threshold, demonstrating that the coating maintains good adhesion to the substrate during high-temperature stages, thereby ensuring effective protection of the TC4 alloy surface.

Fig. 5(a) presents the TG-DSC curves of the glass lubricant. Thermogravimetric analysis results indicate that the total mass loss of the sample from room temperature to 400 °C is approximately 4.55%, which can be divided into two main stages. The mass decrease in the first stage (25–200 °C) is primarily attributed to the evaporation and removal of moisture from the sample. The second stage (200–400 °C) corresponds to the breakdown of methylcellulose polymer chains, pyrolysis, and the release of volatile small-molecular products. During this process, the methylcellulose molecular chains undergo an oxidation reaction with oxygen, generating products such as CO_2 and H_2O , accompanied by significant heat release. As a result, a distinct exothermic peak is observed in the DSC curve at approximately 352 °C. After 400 °C, the TG curve stabilizes, indicating the complete decomposition of methylcellulose, with the remaining components consisting of the borosilicate glass system and no further significant mass changes. A relatively strong exothermic peak appears in the DSC curve near 734 °C, which may correspond to the crystallization of the borosilicate glass; however, this requires confirmation by X-ray Diffraction.

Fig. 5(b) illustrates the viscosity–temperature curve of the glass lubricant. In this study, the temperature-dependent viscosity of the glass lubricant at elevated temperatures was experimentally characterized and analyzed using the Vogel–Fulcher–Tammann (VFT) model [44], [45]. It is well-established that the VFT equation effectively fits viscosity data spanning approximately 10 orders of magnitude (typically from 10^3 to $10^{13} \text{ Pa} \cdot \text{s}$) within the temperature range where rapid structural rearrangements occur in glassy systems [46]. The mathematical expression of the VFT model is given by:

$$\log \eta = A + \frac{B}{T - T_0} \quad (1)$$

where η is the viscosity (unit: $\text{Pa} \cdot \text{s}$), T is the temperature (unit: K), and A , B , and T_0 are temperature-independent fitting parameters. The following VFT equation was applied to fit the experimental data across the glassy, supercooled liquid, and melt

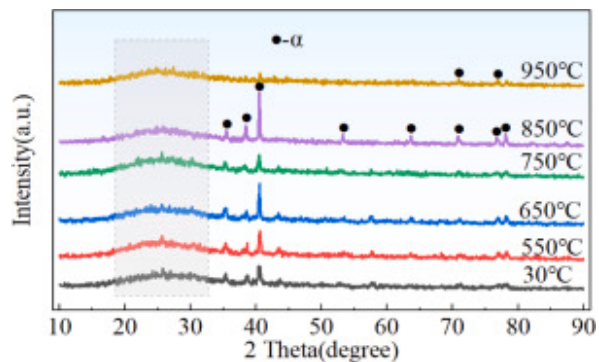
regions:

$$\log \eta = -1.45 + \frac{2322.54}{T-636.41} \quad (2)$$

The adjusted R^2 value of the fitting is 0.99, and the results align well with the viscosity corresponding to the glass transition temperature (T_g). As shown in the figure, the viscosity of the glass lubricant exhibits an exponential decrease with increasing temperature. This characteristic plays a pivotal role in governing the diffusion behavior of the glass melt.

The melting kinetics of glass lubricants exhibit significant dependence on particle dimensions due to their inherent size-specific surface area relationships. Finer particles ($< 1 \mu\text{m}$) demonstrate accelerated melting rates owing to their enhanced specific surface area ($\text{SSA} = 3.6 \text{ m}^2/\text{g}$, which facilitates superior heat transfer efficiency and interfacial reactivity with surrounding media. In contrast, particles with a larger particle size may take a longer time to completely melt. Therefore, we used ImageJ to statistically analyze particle size distribution of the glass particles in the glass lubricant, as shown in Fig. 6(a). The particle size distribution is in the range of 0.28–2.159 μm . The particle sizes of most particles are concentrated below 1.5 μm , and the median d50 of the particle size distribution is approximately 0.625 μm (the result obtained by fitting 120 powder particles). The fitting by the Origin software indicates that the particle sizes of the glass particles are relatively uniform.

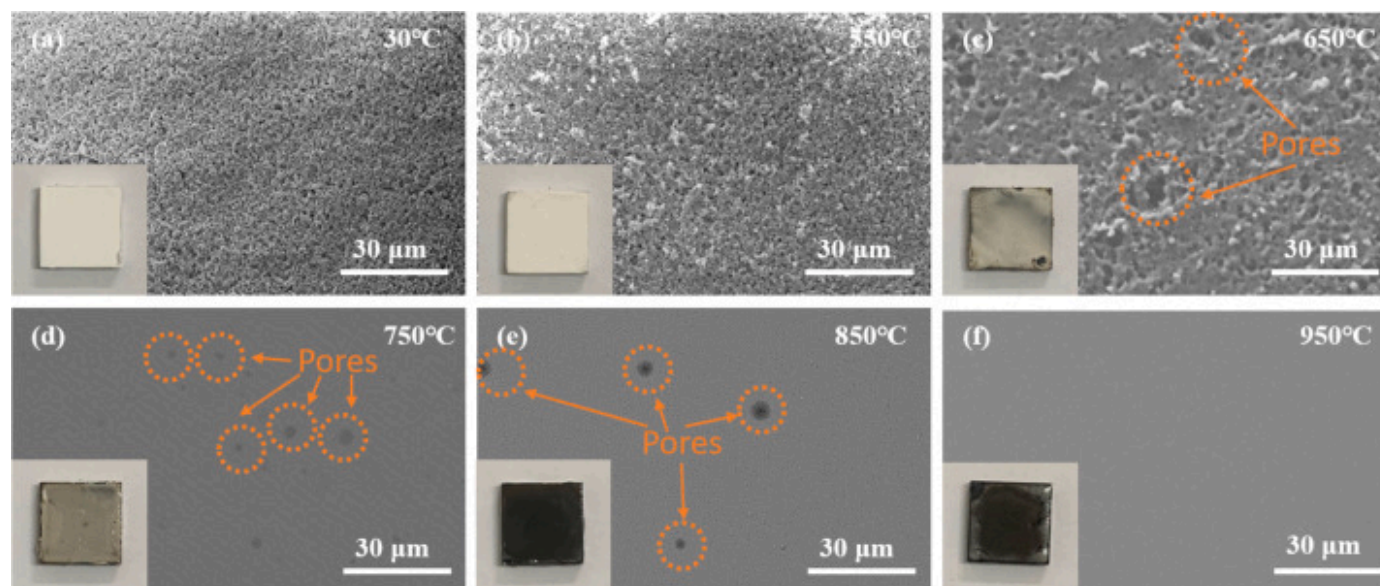
To investigate the dynamic evolution of glass lubricants during thermal processing, linear shrinkage rates of the glass lubricant after heat treatment were systematically analyzed, as illustrated in Fig. 6(b). At 550 °C, initial shrinkage ($5.34 \pm 0.41\%$ mean) occurred primarily through dehydration and pyrolysis of residual moisture and carboxymethyl cellulose. As the temperature increased to 650 °C, the average shrinkage rate reached 9.34%, primarily due to the initial melting of glass particle surfaces, which formed a liquid phase that facilitated enhanced interparticle contact. This process densified the structure through liquid-phase sintering facilitated particle rearrangement and gas expulsion through the evolving porous structure, collectively contributing to the reduction in overall volume. When the temperature rises to the range of 750 °C–950 °C, the glass particles in the glass lubricant have melted and flowed, leading to the shrinkage of the lubricant. At this stage, the sintering of the lubricant is approaching completion, and the volume shrinkage tends to be stable around 5%. The melting and diffusion of the particles have almost reached saturation, and the rate of pore elimination has significantly slowed down. The majority of pores have been eliminated, and the volume of the sample no longer changes significantly, the minor changes mainly attributed to the removal of the gas bubbles within the lubricant.



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Fig. 7. XRD patterns of glass lubricants with increasing Temperature.



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Fig. 8. Macroscopic and Microscopic Morphological Evolution of Glass Lubricant Surface with Increasing Temperature.

3.3. Evolution of phase and morphology of glass lubricants with temperature

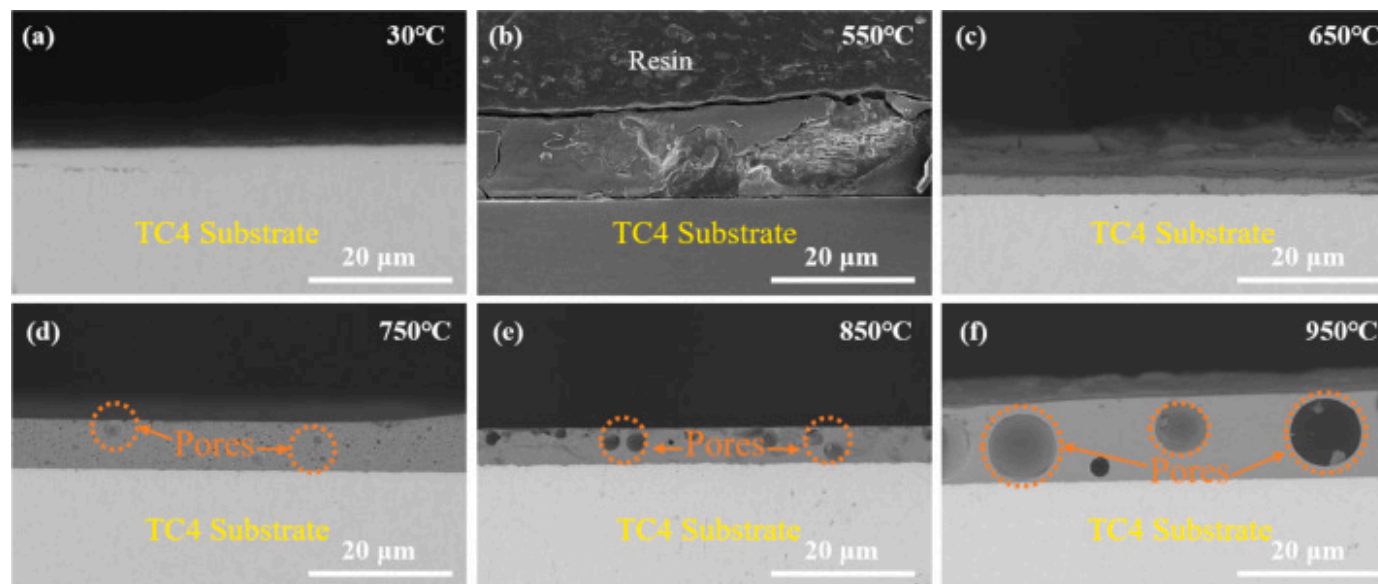
We conducted XRD analysis on glass-coated TC4 alloy substrates across varying temperatures (Fig. 7) to investigate the phase evolution of the lubricant under thermal exposure. The patterns consistently exhibited broad “bun-shaped” humps within the $15\text{--}35^\circ 2\theta$ range, characteristic of amorphous glass structures, alongside well-defined TC4 alloy substrate diffraction peaks. Notably, no oxide-related diffraction peaks were observed throughout the heat treatment process, which demonstrates that the glass lubricant not only exhibits an anti-oxidation effect on the TC4 alloy substrate but also maintains excellent thermal stability.

In order to study the changes in the macroscopic morphology of the samples with temperature, we collected optical photographs of the samples coated with the glass lubricant as the temperature varied, as shown in Fig. 8. When the sample was heated to 550°C , there was no significant difference in the optical photograph of the sample compared with that at room temperature. At 650°C , partial melting occurred in the glass lubricant while retaining numerous unmelted white glass particles. As the temperature increased to 750°C , enhanced melting reduced the amount of unmelted white glass. Upon reaching 850°C to 950°C , complete melting of the glass lubricant was achieved, resulting in a fully transparent and glossy surface.

Fig. 8 displays the as-prepared glass lubricant and its surface microstructural evolution after heating to 550°C , 650°C , 750°C , 850°C , and 950°C with 30-minute holding periods. The scanning electron microscopy image in Fig. 8(a) reveals that the original lubricant surface consists of spread glass particles. At 550°C (Fig. 8(b)), the lubricant's microstructure showed no significant changes, indicating the glass particles remained unmelted. Partial softening and melting of glass particles occurred at 650°C (Fig. 8(c)), forming a porous morphology. Further heating to 750°C (Fig. 8(d)) induced complete particle melting into a continuous film containing numerous fine pores with diameters ranging $0.53\text{--}2.13\text{ }\mu\text{m}$ as quantified by Image J analysis. At 850°C (Fig. 8(e)), pore density decreased while pore diameters expanded to approximately $2.38\text{--}5.02\text{ }\mu\text{m}$. Ultimately, the lubricant achieved full densification at 950°C (Fig. 8(f)), exhibiting a smooth vitreous surface with virtually eliminated porosity.

Based on the aforementioned surface microstructural observations of the glass lubricant, we subsequently investigated the cross-sectional morphology evolution under corresponding thermal conditions. Fig. 9 illustrates the as-prepared glass lubricant and its cross-sectional microstructural changes after 30-minute isothermal treatments at 550°C , 650°C , 750°C , 850°C , and 950°C , respectively. The SEM image of the original lubricant cross-section (Fig. 9(a)) reveals partial delamination from the TC4 alloy substrate during sample preparation due to inadequate bonding in the unsintered state. At 550°C (Fig. 9(b)), weak interfacial adhesion is observed between the lubricant and both the TC4 alloy substrate and resin matrix. This is accompanied by heterogeneous features, such as elongated slit-like regions and floral surface textures, which are likely transitional structures

formed during the initial heating process. Improved substrate bonding emerges at 650 °C (Fig. 9(c)) with glass particle melting, though irregularly shaped pores develop. Heating to 750 °C (Fig. 9(d)) generates numerous shallow circular bubbles (0.8–6.76 μm diameter) containing fine glass particulates. At 850 °C (Fig. 9(e)), bubbles density decreases while diameters expand (5.17–28.2 μm), exhibiting both smooth and particle-retaining inner surfaces. Final treatment at 950 °C (Fig. 9(f)) results in significantly enlarged bubbles (25.67–98.03 μm diameter) with enhanced depth and complete surface smoothing, indicating advanced sintering stages. The apparent thickening at 950 °C is mainly attributed to an edge-accumulation effect: under low-viscosity, high-mobility conditions, the molten material undergoes radial migration along the surface and, near the specimen boundary, becomes pinned at the contact line and constrained by the edge geometry, thereby forming a thicker rim layer with more pronounced edge convergence and rim-building behavior (Fig. 8(f), Fig. 9(f))



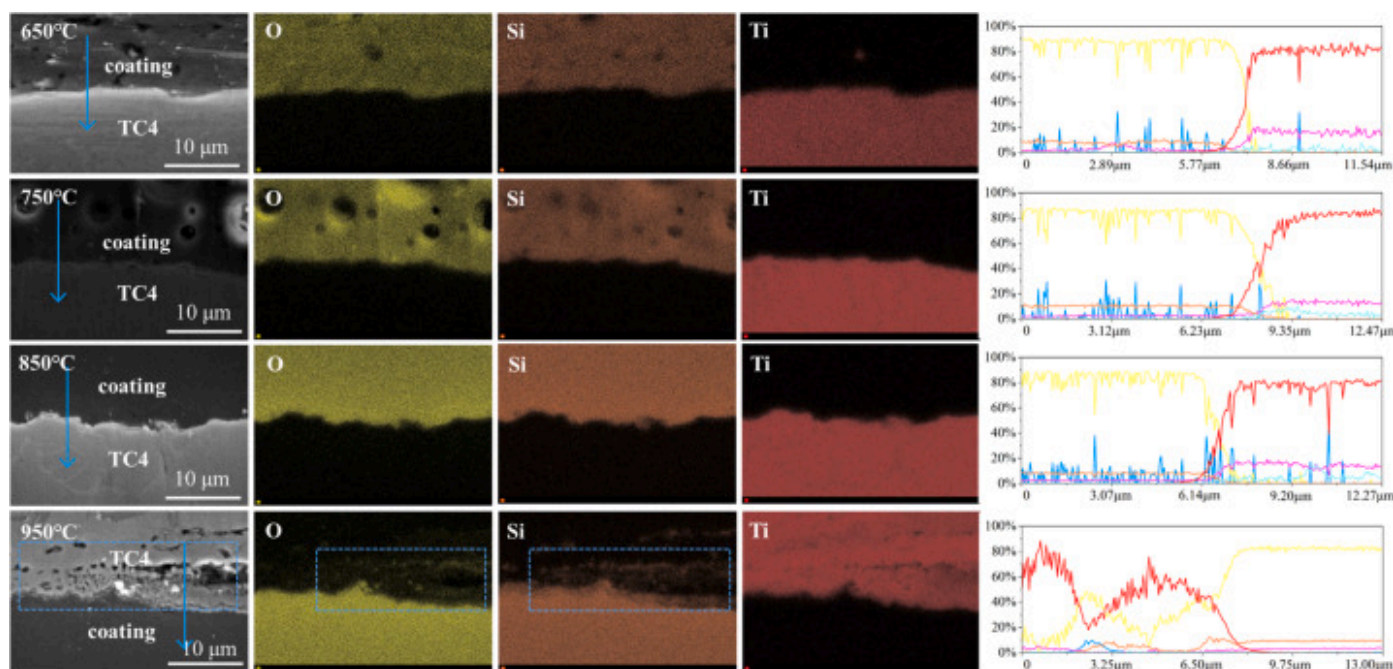
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Fig. 9. Microscopic morphological evolution of glass lubricant cross-section with increasing temperature. (The image magnification is consistent, and the difference in coating thickness is caused by coating shrinkage or edge effects.).

3.4. Interface analysis of glass lubricant/TC4 alloy substrate

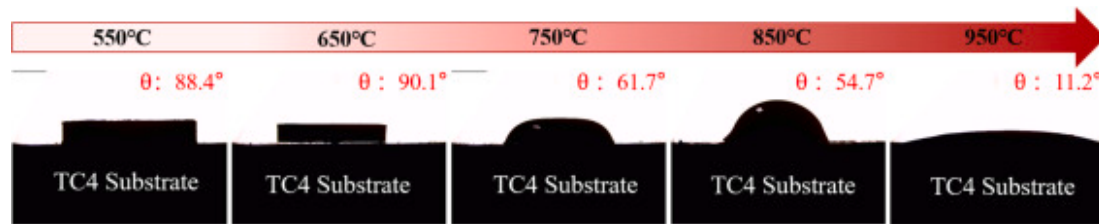
Fig. 10 presents the elemental distribution at the interface of the glass lubricant after heat treatment at 650 °C, 750 °C, 850 °C, and 950 °C for 30 min. EDS analysis, including elemental mapping and line scanning, revealed the following: at 650 °C, 750 °C, and 850 °C, oxygen (O) and silicon (Si) were predominantly concentrated within the lubricant, while titanium (Ti) remained largely confined to the TC4 alloy substrate. Interdiffusion across the interface was limited, resulting in a relatively well-defined boundary between the lubricant and the substrate, suggesting minimal interdiffusion under these conditions. In contrast, at 950 °C, significant interdiffusion of O, Si, and Ti was observed, particularly within the dashed region. Elemental maps showed mutual penetration of these elements near the interface, and line scans indicated broader interdiffusion zones with smoother concentration gradients. This suggests substantial interdiffusion and the formation of a reaction layer at this temperature. The interdiffusion between Si and Ti likely promotes the formation of Ti_5Si_3 , $\text{Ti}_6\text{Si}_2\text{B}$ [15], [47]. The EDS results confirm that interdiffusion, particularly between Si and Ti.



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Fig. 10. Elemental Analysis of Glass Lubricant Interface with Increasing Temperature (Yellow represents the O element, orange represents the Si element, and red represents the Ti element, blue represents the B element, purple represents the Al element, and light green represents the V element.).



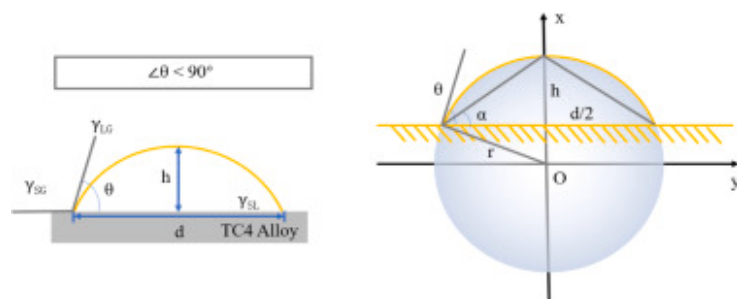
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Fig. 11. High-temperature wetting evolution of glass lubricant on TC4 alloy substrate with increasing temperature.

3.5. Evolution of high-temperature wettability of glass lubricants on TC4 alloy substrate

The high-temperature wettability of borosilicate glass lubricants on TC4 alloy substrates was systematically investigated using the sessile drop method with a contact angle analyzer, as illustrated in Fig. 11. As the temperature rose, the cylindrical glass body gradually spread, with the process divided into four stages. Stage I (Room Temperature–650 °C) exhibited volumetric contraction while maintaining geometry, attributed to particle softening/melting-induced void consolidation and decomposition and removal of methylcellulose. Excessive internal stresses generated during densification caused interfacial delamination from the TC4 alloy substrate (Fig. 11(650 °C)). Stage II (650–750 °C) manifested geometric restructuring through upper-edge rounding, converting sharp corners into smooth curvatures with a measured contact angle of 61.7° (Fig. 11(750 °C)). Stage III (750–850 °C) demonstrated enhanced interfacial compatibility, reducing the contact angle to 54.7° through the wetting effect between the borosilicate glass and the TC4 alloy substrate (Fig. 11(850 °C)). Stage IV (850–950 °C) demonstrated that the viscous glass continued to spread. As the temperature increased, the viscosity of the glass decreased, resulting in enhanced fluidity and a broader spreading area. This process led to a reduction in the macroscopic contact angle to 11.2° (Fig. 11(950 °C)).



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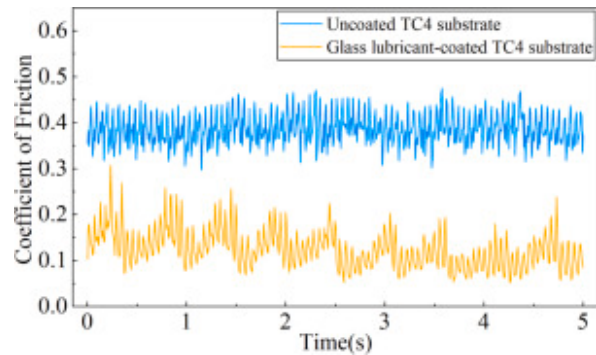
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Fig. 12. Mathematical model of contact angle of glass melt.

3.6. Analysis of the friction properties of glass lubricants

Fig. 13 compares the time-dependent friction coefficients at 900 °C for TC4 alloy with and without a glass-lubricant coating. The uncoated TC4 alloy maintains a consistently high friction level throughout the test, with the coefficient mainly in the range of 0.36–0.41 and continuous high-frequency fluctuations about an apparent steady plateau; transient maxima approach 0.48, indicating pronounced frictional instability. In contrast, the glass-coated TC4 alloy exhibits a substantially reduced coefficient, predominantly 0.09–0.15, together with markedly attenuated fluctuations. Only intermittent spikes are observed, which are attributed to momentary disturbances of the molten film followed by melt backflow and rapid self-healing.

The friction reduction arises from the formation of a continuous molten phase at high temperature that wets and spreads over the substrate, establishing a stable “molten isolation film”. This film converts direct metal–metal contact into indirect metal–glass melt–counterface contact, thereby providing interfacial isolation and suppressing oxidation-assisted adhesion between TC4 alloy and the die. Moreover, the amorphous glass network in the molten state can undergo structural rearrangement; under shear, ions/ionic clusters experience relative displacement that facilitates interfacial shear slippage, reducing the effective shear resistance. Owing to its high flowability and spreadability, the melt dynamically replenishes local film discontinuities, fills wear grooves, and enables localized self-repair under reciprocating shear and compressive loading, maintaining film continuity and load-bearing stability and ultimately delivering sustained high-temperature anti-adhesion lubrication.



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Fig. 13. Coefficient of friction at 900 °C.

4. Discussion

4.1. Wettability analysis of glass lubricant on TC4 alloy substrate

To elucidate the spreading–wetting process of vitreous material on TC4 alloy surfaces, a mathematical model was employed to quantify the contact angle (θ), as depicted in Fig. 12. The analysis required measurements of the melt's diffusion dimensions, specifically the spread height (h) and base diameter (d) of the hemispherical melt. Geometrically, the tangent trigonometric relationship ($\tan \angle \alpha = \frac{2h}{d}$) determined the angle α , while the macroscopic contact angle θ was derived through $\theta = 2 \cdot \arctan \left(\frac{2h}{d} \right)$.

The contact angle θ quantifies the wettability of molten glass on TC4 alloy substrates, with the wetting mechanism governed by the Young–Laplace equation [48], [49], [50]. The fundamental relationship is expressed as:

$$\gamma_{SL} + \gamma_{LG} \cos \theta = \gamma_{SG} \quad (3)$$

$$F = \gamma_{LG} \cos \theta = \gamma_{SG} - \gamma_{SL} \quad (4)$$

where γ denotes interfacial tension (subscripts S , L , G representing solid, liquid, and gas phases respectively) and F corresponds to wetting tension. This formulation demonstrates that the equilibrium contact angle θ is directly determined by the

thermodynamic balance between three interfacial tensions: solid–liquid (γ_{SL}), liquid–gas (γ_{LG}), and solid–gas (γ_{SG}). Specifically, lower γ_{SL} and higher γ_{SG} synergistically enhance wettability by reducing θ , while elevated γ_{LG} values counteract spreading through increased surface energy. The Young–Laplace framework thus provides critical insights into designing glass lubricants through interfacial energy manipulation. To elucidate the wettability evolution of the borosilicate glass lubricant on TC4 alloy substrates governed by the Young–Laplace equation (as illustrated in Fig. 12), the following mechanistic interpretation is proposed. Upon temperature elevation, the softened glass melt infiltrates TC4 alloy surface grooves via capillary forces, filling surface scratches and amplifying the anchoring effect induced by substrate roughness. This enhanced physical bonding increases the effective solid–liquid contact area, thereby reducing the apparent contact angle. Below 850 °C, the primary drivers of contact angle reduction are:

1. Solid surface energy enhancement. Accelerated oxidation of TC4 alloy above 650 °C (Section 3.1) generates a dense oxide layer ($\text{TiO}_2/\text{Al}_2\text{O}_3$), elevating surface energy and intensifying solid–liquid interfacial tension. This increases the driving force for wetting ($\Delta\gamma = \gamma_{SG} - \gamma_{SL}$), lowering θ according to:

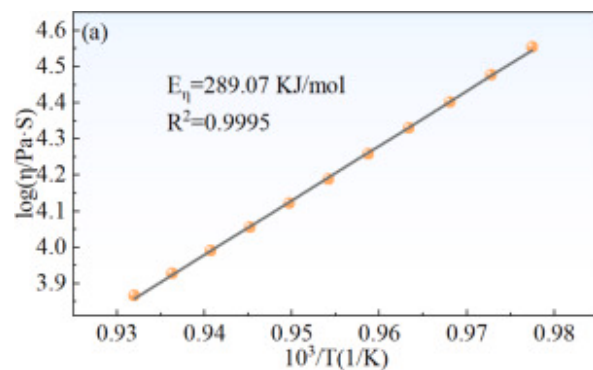
$$\cos \theta = \frac{\gamma_{SG} - \gamma_{SL}}{\gamma_{LG}} \quad (5)$$

2. Substrate roughness amplification. Due to the oxidation of the TC4 alloy surface, the roughness of the substrate increases (as shown in Fig. 3). The wetting of the molten glass lubricant on the rough TC4 alloy surface follows the Wenzel wetting mode. According to the Wenzel equation:

$$\cos \theta_{\text{rough}} = r \cdot \cos \theta_{\text{smooth}} \quad (6)$$

where r is the roughness factor. The surface roughness of TC4 alloy amplifies its inherent wettability. As the melt infiltrates into the microscopic depressions on the solid surface, the contact area between the liquid and solid increases, thereby enhancing the wetting behavior and reducing the contact angle of the liquid droplet.

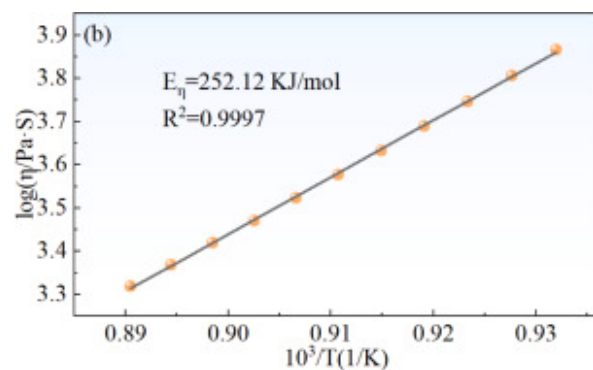
Above 850 °C, contact angle reduction primarily results from melt surface tension diminution (γ_{LG}). Glass viscosity decreases exponentially, weakening intermolecular forces and lowering γ_{LG} . This reduction in γ_{LG} directly decreases θ through the Young–Laplace relationship, enabling complete spreading ($\theta=11.2^\circ$) at 950 °C.



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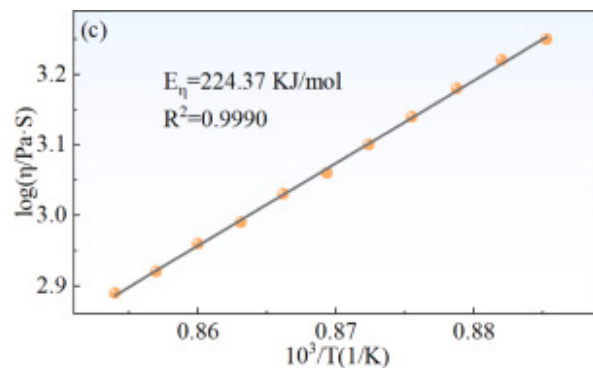
(a)..



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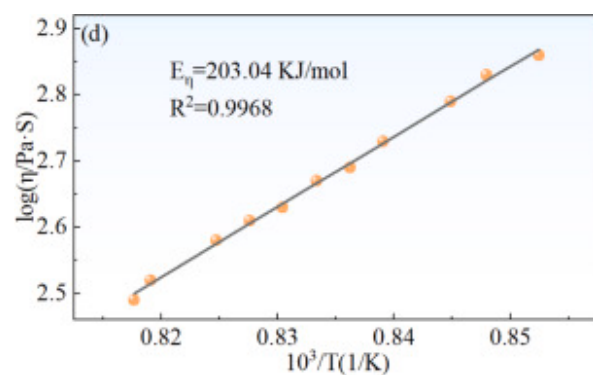
(b)..



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(c)..

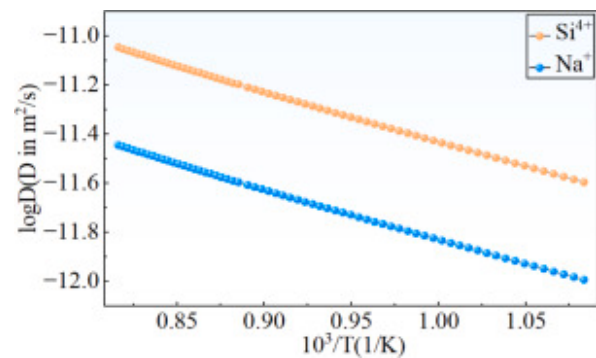


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(d)..

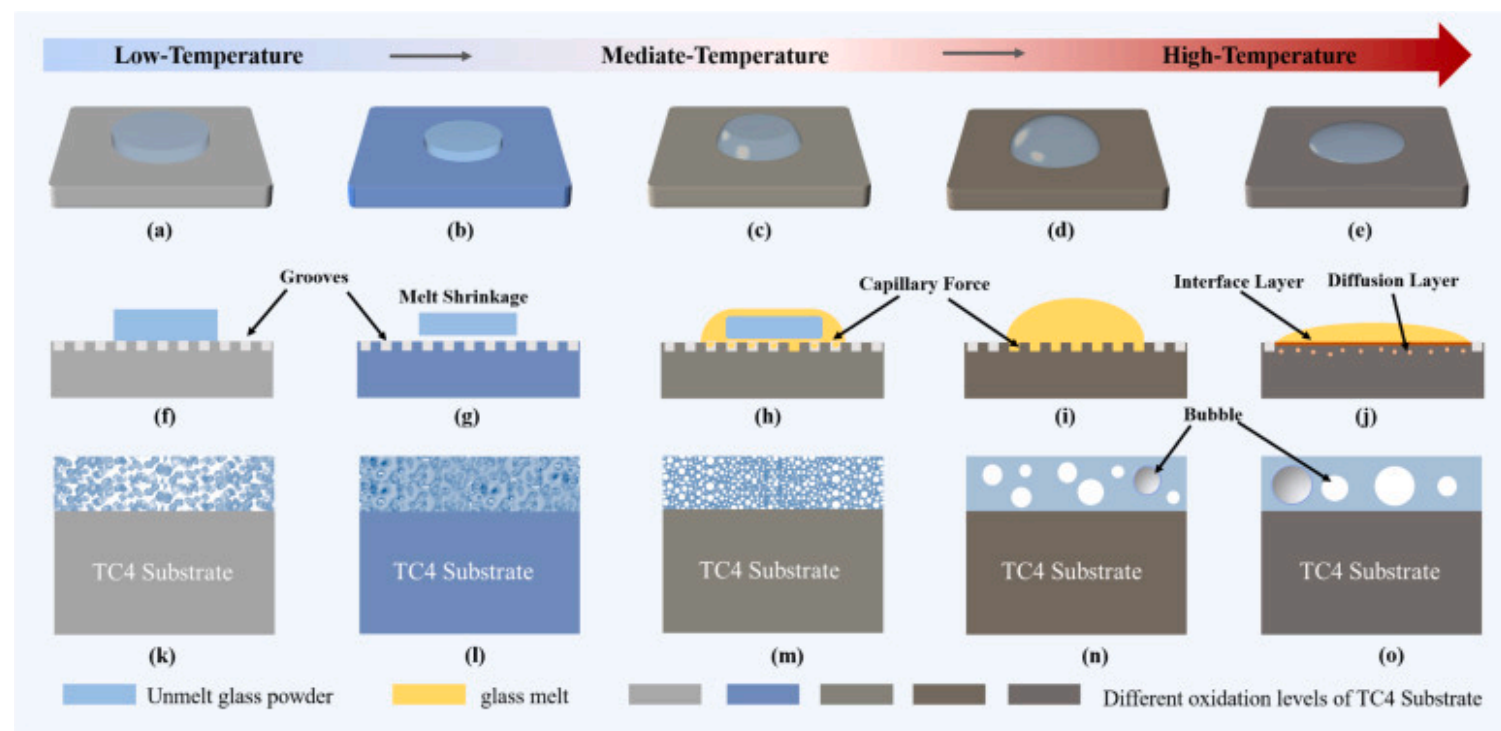
Fig. 14. Viscosity–temperature reciprocity of glass Lubricants:(a) 750–800 °C,(b) 800–850 °C,(c) 850–900 °C,(d) 900–950 °C.



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Fig. 15. The diffusion coefficient of ions in glass lubricants.



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Fig. 16. Schematic diagram of morphological evolution and wetting behavior of glass lubricant on TC4 alloy substrate with increasing temperature: (a)–(e) wetting of glass lubricant on TC4 alloy substrate; (f)–(j) cross-sectional view of glass lubricant wetting on TC4 alloy substrate; (k)–(o) internal structural evolution of glass lubricant wetting on TC4 alloy substrate.

4.2. Transport kinetics of glass lubricants on TC4 alloy surfaces

Macroscopically, the diffusion of glass melt can be interpreted by the spreading amount S of glass on the TC4 alloy surface, which is closely related to the spreading diameter d . When the glass melt starts to spread and diffuse on the TC4 alloy surface, the diffusion rate can be characterized by the diameter d of the melt spreading on the TC4 alloy surface. A faster change rate of the d -value indicates a quicker spreading rate, while a larger d -value signifies a greater spreading amount. The measured spreading diameters were 12 mm at 550 °C, 9.71 mm at 650 °C, 9.96 mm at 750 °C, 9.6 mm at 850 °C, and 17.66 mm at 950 °C. Within the lower temperature range of 750 °C–850 °C, the d -value shows limited variation, decreasing slightly from 9.96 mm to 9.6 mm, which corresponds to a variation rate of approximately $-3.6 \mu\text{m}/^\circ\text{C}$, demonstrating almost no spreading. In contrast, a significant enhancement in spreading kinetics occurs at elevated temperatures (850 °C–950 °C), where the d -value increases markedly from 9.6 mm to 17.66 mm, with a variation rate of approximately $80.6 \mu\text{m}/^\circ\text{C}$, indicating rapid expansion of the glass melt on the TC4 alloy substrate.

Microscopically, the transport dynamics of glass melt primarily involve two “driving forces” for mass transport: viscosity (η) and diffusion (D) [51], [52], [53]. Under the influence of external forces (such as gravity and thermal stress), adjacent ions and ionic clusters within the molten glass experience relative motion, fill microstructural units to occupy structural vacancies and thereby inducing viscous flow within the glass matrix. The viscous forces between glass melts originate from this viscous flow, with viscosity serving as a quantitative measure of these forces. Consequently, the transport behavior of glass melts can be systematically analyzed through viscosity characterization. Notably, The viscosity of the glass melt follows the Arrhenius equation within a narrow range (below 10^3), with the corresponding viscous activation energy being constant [54]. The viscosity (η) of the glass lubricant can be expressed as [55], [56]:

$$\eta(T) = \eta_0 \exp\left(\frac{E_\eta}{RT}\right) \quad (7)$$

where η_0 is the pre-exponential factor, $\eta(T)$ is the viscosity at temperature T , E_η is the viscous activation energy (unit: J/mol), R

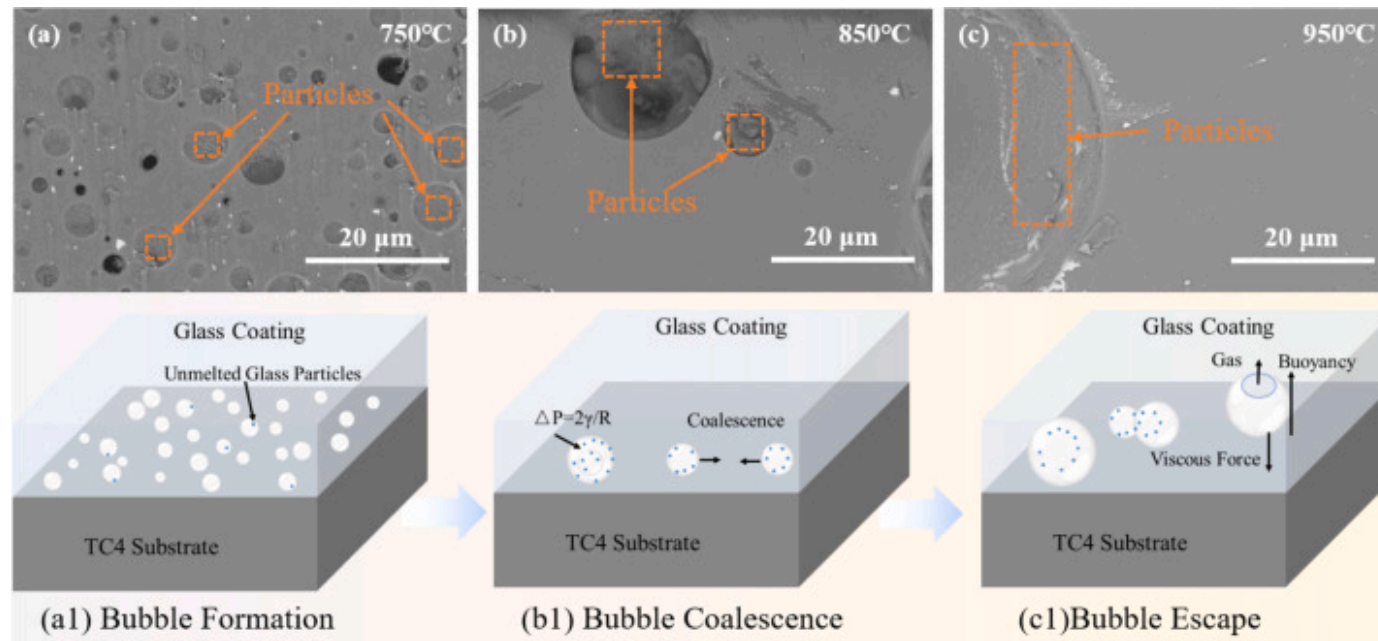
is the gas constant ($8.314 \text{ J}/(\text{mol K})$), and T is the thermodynamic temperature (unit: K). As revealed in Section 3.5, the glass lubricant begins to spread and diffuse at 750°C . Therefore, this work investigates the spreading and diffusion behavior of the glass lubricant within the temperature range of $750\text{--}950^\circ\text{C}$. The temperature range was divided into 50°C intervals, assuming that the viscous activation energy E_η of the glass lubricant remains constant within each interval. This approach resembles a differential process. The piecewise fitting using the Arrhenius equation (Fig. 14) demonstrates that the viscosity of the glass lubricant exhibits a linear relationship with the reciprocal of temperature ($1000/T$), consistent with the Arrhenius equation. The viscous activation energy E_η values for the temperature intervals of $750\text{--}800^\circ\text{C}$, $800\text{--}850^\circ\text{C}$, $850\text{--}900^\circ\text{C}$, and $900\text{--}950^\circ\text{C}$ were determined to be 289.07 kJ/mol , 252.12 kJ/mol , 224.37 kJ/mol , and 203.04 kJ/mol , respectively. These results indicate a gradual decrease in E_η with increasing temperature. This trend can be attributed to enhanced thermal motion of molecules at elevated temperatures, which weakens intermolecular interactions and reduces the energy barrier required for viscous flow. Consequently, the relative movement between molecules becomes more facile, facilitating the spreading of the glass lubricant on the TC4 alloy substrate. The diffusion process in molten glass constitutes a critical mass transport mechanism. The “diffusion driving force” is fundamentally associated with the fundamental displacement of atoms at the atomic scale. For high-viscosity fluid dynamics, the Stokes–Einstein equation establishes a quantitative correlation between diffusion and viscosity, expressed as [57], [58], [59]:

$$D = \frac{k_B T}{6\pi\eta r} \quad (8)$$

where D is the diffusion coefficient of the particle (unit: m^2/s); k_B denotes the Boltzmann constant ($1.38 \times 10^{-23} \text{ J/K}$; T represents the absolute temperature (unit: K); η is the fluid viscosity (unit: Pa s); and r is the particle radius (unit: m).

Based on Arrhenius viscosity equation and the Stokes–Einstein equation, the diffusion coefficients of Si^{4+} and Na^+ ions in the system were calculated. As shown in Fig. 15, the calculated results exhibit agreement with literature data [60], [61], [62], [63]. It is important to note that the theoretical calculations here focus solely on the diffusion coefficients of individual ions, whereas actual diffusion processes may be influenced by additional factors, such as interactions between particles or particle clusters and variations in melt composition. Specifically, silicon may not diffuse as isolated Si^{4+} ions but rather in the form of $[\text{SiO}_4]$ units. The diffusion of such units requires breaking multiple Si–O bonds and restructuring the silicate network, which involves extremely high activation energy and results in a significantly slower diffusion rate. Nevertheless, the calculated values still provide valuable insights into the diffusion behavior. With increasing temperature, the diffusion coefficients of both Si^{4+} and Na^+ ions increase, thereby enhancing the diffusion of the glass melt. When the viscosity stabilizes, temperature becomes the dominant

factor governing diffusion, and the diffusion coefficient rises with elevated temperatures. In conclusion, the synergistic effects of viscosity and diffusion collectively drive the spreading of the glass melt on the TC4 alloy substrate.



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Fig. 17. (a), (b), and (c) are the SEM images of the bubbles at temperatures of 750 °C, 850 °C, and 950 °C respectively and (a1), (b1), and (c1) are schematic diagrams illustrating the behavior of bubbles in the glass lubricant.

4.3. Formation mechanism of glass lubricants on TC4 alloy substrate

As depicted in the schematic illustration of glass film formation on TC4 alloy surfaces (Fig. 16), the glass particles within the lubricant progressively soften and melt as the temperature increases, resulting in a rough and porous surface structure attributed to disparities in particle size (Fig. 16(l)). As the temperature continues to rise, the glass powders melt and spread into a continuous film on the TC4 alloy substrate, forming a smooth and uniform surface. However, numerous microscopic pores are generated within the glass film (Fig. 16(m)). These pores primarily originate from gases trapped between the glass particles and bubbles formed by gases released from the decomposition of dimethyl cellulose. A portion of the internal bubbles coalesce and

expand to form larger cavities, while others escape from the glass film surface as the temperature continues to increase (Fig. 16(n)–(o)).

From a microstructural perspective, the glass film can be conceptualized as a capillary-like medium, where the melt redistribution is primarily driven by capillary pressure arising from the surface/interfacial tension of the glass melt [64](Fig. 16(f)–(j)). When the molten glass resides within the pores of the film, interactions occur between the melt and the pore walls. Due to unbalanced intermolecular forces acting on the surface molecules of the melt, a surface tension directed inward is generated. Within the confined geometry of the pores, this surface tension induces a tendency for the melt to contract toward its center, thereby creating capillary forces that drive melt flow and bubble growth. The spontaneous motion of the melt in the capillary originates from surface diffusion of atoms at the meniscus/capillary interface boundary [65]. This leads to a slight increase in meniscus curvature and correspondingly elevated surface tension. The higher surface tension tends to restore the original curvature of the meniscus. Consequently, the melt undergoes viscous flow from the central region of the meniscus to new positions, facilitating bubble coalescence and expansion. As bubbles evolve, the capillary pressure difference across the bubble interface is governed by the Laplace equation [66], i.e., $\Delta P = 2\gamma/R$, where γ is the surface/interfacial tension and R is the bubble radius. This relation indicates that smaller bubbles (smaller R) possess a higher internal pressure, whereas ΔP decreases as bubbles grow. Consequently, the pressure imbalance between neighboring bubbles provides a thermodynamic driving force for coarsening and coalescence, whereby gas transfer or bubble merging preferentially increases the size of larger bubbles at the expense of smaller ones. Importantly, surface/interfacial tension does not directly “promote” bubble growth; instead, it tends to minimize the total interfacial area. Therefore, the system can reduce its interfacial energy through bubble coalescence and coarsening, which macroscopically manifests as the emergence of fewer but larger bubbles. When a bubble reaches a critical size where buoyancy exceeds the viscous resistance of the melt, it begins migrating toward the melt surface. During this migration, collisions with other bubbles may lead to coalescence, further increasing bubble volume. Additionally, elevated temperatures reduce the solubility of dissolved gases in the glass melt, causing partial bubble escape from the film surface (Fig. 17) [67]. Notably, unmelted glass particles are observed to accumulate around bubbles (Fig. 17). This phenomenon can be attributed to the adsorption of nearby unmelted particles onto bubble surfaces under the combined effects of buoyancy and surface tension during bubble formation.

5. Conclusion

This study systematically investigates the high-temperature wetting behavior and film-forming mechanism of a borosilicate glass lubricant on TC4 alloy substrates. The results demonstrate that the glass lubricant exhibits excellent wettability and film-forming capability under high-temperature conditions. As the temperature increases from 550 °C to 950 °C, the lubricant undergoes a dynamic evolution process of “softening–melting–spreading”, ultimately forming a continuous glass film with a contact angle as low as 11.2° at 950 °C, indicating superior spreading characteristics. Further analysis reveals that the wetting mechanism follows the Wenzel mode, induced by the combined effects of surface roughness and the presence of a TiO₂/Al₂O₃ layer. The film formation is governed by the synergistic influence of high-temperature viscosity reduction, diffusion kinetics, and capillary forces: at the initial stage, capillary action drives the molten glass to fill surface grooves and interfacial voids; at elevated temperatures, decreased viscosity and enhanced diffusion of ions and ion clusters accelerate the formation of a dense and continuous glass film. The findings provide a theoretical foundation for the development of high-performance glass lubricants for high-temperature protection of TC4 alloy.

CRedit authorship contribution statement

Meijing Chen: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Xin Tian:** Writing – review & editing, Validation, Investigation, Formal analysis. **Hengjun Luo:** Validation, Investigation. **Detao Zeng:** Investigation. **Ying Liu:** Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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